

Sarthik Mandarapu | Software Engineer

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SUMMARY

Results-driven Software Engineer with **4+ years** of experience building scalable backend systems, **REST APIs**, and AI-integrated solutions using **Java, Python, Spring Boot**, and **AWS**. Proven track record in **CI/CD automation**, cloud deployments, and data-driven development. Skilled in integrating **ML models**, optimizing system performance, and securing applications with **OAuth2 and JWT**. Adept at **Agile** collaboration, test automation, and microservice architecture, delivering high-impact solutions across **healthcare, finance, and enterprise domains**.

TECHNICAL SKILLS

Languages & Scripting: Java (JDK 8/11/17+), Python (Django, Flask), JavaScript (ES6+), TypeScript, SQL, C# (.NET)

Backend Technologies: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, Apache Kafka,

Frontend Technologies: Angular, React.js, HTML5, CSS3, Bootstrap, Material UI, Axios

Database Technologies: MySQL, PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, S3, IAM, RDS, Lambda), Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines

Testing & Monitoring: JUnit, Mockito, REST Assured, Postman, Swagger (Open API), Prometheus, Grafana, Git

AI/ML Integration: MLFlow, Prompt Engineering, OpenAI API, Hugging Face, TensorFlow, scikit-learn

Security Practices: OAuth2, JWT, Spring Security, SSL/TLS, HTTPS, API Key Authentication, Role-Based Access Control (RBAC)

Build & Version Control: Maven, Gradle, Git, GitHub, GitLab

Tools & Platforms: IntelliJ IDEA, VS Code, JIRA, Confluence, Agile/Scrum

Operating System: Windows Server, Linux/Unix, Mac OS

PROFESSIONAL EXPERIENCE

Dell Technologies, TX, USA

Jul 2024 – Present

Software Engineer

- Reduced authentication issues by **30%** through **secure integration** of **OAuth2** and **JWT** in **Spring Boot** microservices, enhancing security posture and reducing user access-related errors across distributed environments.
- Integrated **OpenAI GPT APIs** for chatbot automation in internal tools, streamlining support workflows and reducing average ticket resolution time by **40%** in the customer service knowledge base platform.
- Developed robust **Python pipelines** using **Django** to transform and normalize complex enterprise datasets, significantly increasing the efficiency of machine learning model ingestion workflows and **reducing data wrangling time by over 35%** for crucial experimentation and analysis.
- Deployed Hugging Face sentiment classification models via Flask REST APIs, enabling real-time analysis of user reviews and improving product feedback insights for stakeholder decision-making.
- Automated CI/CD deployments using **Docker** and **GitHub Actions** on **AWS EC2/RDS**, **decreasing release cycles by 55%**.
- Documented **REST APIs** using **Swagger** and OpenAPI standards, improving collaboration with frontend teams and reducing integration bugs by **25%** through clearer versioning and request-response structure clarity.
- Achieved **85%** test coverage with **JUnit** and **Mockito**, leading to faster debugging and a significant reduction in post-release defects.
- Deployed containerized microservices using **Docker** and **Kubernetes** on **AWS** and **Azure**, improving scalability and reducing deployment issues by **40%**.
- Integrated **Prometheus** and **Grafana** for live metrics monitoring and alerting, **reducing mean time to resolution (MTTR) by 30%**.
- Designed backend modules using **Spring Boot** and **Ruby on Rails** to support AI-driven workflows and internal tooling.
- Implemented and tracked ML model lifecycles using **MLFlow**, enabling reproducible training, version control, and experiment tracking.

Dell Technologies, TX, USA

Jan 2024 – Jun 2024

Software Engineer Intern

- Designed **REST APIs** with **PostgreSQL**, **boosting dashboard responsiveness by 28%** for internal auditing applications.
- **Automated test coverage** using **JUnit** and **Mockito**, increasing reliability of intern-owned modules by **45%** and catching deployment bugs early during staging in two internal product lines.
- **Deployed containerized microservices** using **Docker** on **AWS EC2**, improving deployment efficiency and reducing manual release errors by **50%** under the mentorship of DevOps team leads.
- Integrated **OpenAI API** into the **sandbox feature**, **automating responses** and boosting stakeholder feedback by **33%**.

Hexaware Technologies, India

Sep 2020 – Jul 2022

Software Engineer

- Created **RESTful APIs** using **Spring MVC**, **Hibernate**, and **PostgreSQL**, streamlining backend workflows and improving response accuracy for financial transaction modules handling high-volume user interactions.
- Created and deployed cloud infrastructure using **Terraform** on **Azure**, automating the setup and configuration of scalable

environments for analytics applications.

- Integrated Angular components with backend services to enhance UI responsiveness and enable secure, role-based access.
- Designed **Python** utilities for parsing server logs and detecting anomalies, enhancing error traceability, and reducing average incident investigation time by **28%** across development and production pipelines.
- **Increased fraud detection model accuracy** by optimizing **tuning** and **evaluation** tasks using **TensorFlow** and **scikit-learn**.
- Implemented **Redis** caching strategies and query optimizations, reducing API response latency by **40%** and ensuring consistent performance under concurrent traffic scenarios.
- Developed **Jenkins pipelines** with **Maven** and **Docker**, automating build and deployment stages to improve delivery reliability and reduce manual errors during release processes.
- Created API validation scripts using **REST Assured** and **Postman**, shortening **QA cycles** and improving test coverage for backend endpoints in high-priority service modules.

CitiusTech, India
Java Developer

Jan 2020 – Aug 2020

- Optimized **SQL** queries in **MySQL** to reduce load time for analytics dashboards by **40%**, improving usability for doctors and health administrators using real-time clinical decision support tools.
- Implemented authentication with **Spring Security** and **JWT** tokens, ensuring **HIPAA compliance** for medical information systems.
- Created frontend components using **HTML5**, **CSS3**, and **Bootstrap**, improving the responsiveness and usability of admin dashboards used for hospital staff management workflows.
- Built unit tests with JUnit for service layers, achieving 80% code coverage.

EDUCATION

Master's In Computer Science

May 2024

Texas A&M University-Corpus Christi. Texas.

Bachelor of Technology in Computer Science

Aug 2021

Bharath Institute of Higher Education and Research (BIHER), Chennai

PROJECTS

Smart Healthcare Appointment System

Tech Stack: Spring Boot, React.js, MySQL, JWT, Docker, AWS EC2/RDS, Swagger

- Built a secure appointment platform with **Spring Boot microservices**, **JWT**-based login, and a **React.js frontend**; reduced scheduling conflicts by **40%** and supported multi-role access (patient, doctor, admin).
- Deployed Dockerized services on **AWS EC2** with **MySQL (RDS)**, integrated **Swagger** for **API documentation**, and achieved seamless booking flows with service-level validation and real-time slot availability.

Cloud-Based Expense Tracking Platform

Tech Stack: Spring Boot, MongoDB, Redis, React.js, OAuth2, Docker, GitHub Actions

- Developed a personal finance tracker using **Spring Boot microservices** and **MongoDB**; implemented **Redis** caching to reduce **dashboard** load time by **60%** for high-volume user queries.
- Enabled **GitHub/Google login** via **OAuth2** and **CI/CD** pipeline using **GitHub Actions**; deployed modular features and delivered a responsive **React.js dashboard** with category-wise filtering and export option

